Advertising Spend vs Product Sales

Multiple Linear Regression — Analysis, Insights & Recommendations

Assignment: Task 6 · **Date:** 7 October 2026
Analyst: Mohammed Mutee Ruknuddin · **Role:** Data Analyst
Dataset: Advertising Dataset (Kaggle — `ashydv/advertising-dataset`), 200 records × 4 columns
Deliverables: `Advertising_Sales_Prediction_MohammedMuteeRuknuddin.ipynb` + this report

1. Business Scenario & Objective

A consumer goods company promotes its product across three advertising channels — **Television (TV)**, **Radio** and **Newspaper**. Management needs an answer to one question: *how do these channels together influence product sales, and can future sales be predicted from a planned advertising budget?*

This report quantifies the spend→sales relationship using multiple linear regression, forecasts sales for a specific planned budget, measures how wrong the forecast is likely to be, ranks the three channels by effectiveness, and converts the findings into one business action and one technical action.

Units throughout: advertising spends are in thousands of dollars ($'000) per market; `Sales` is in thousands of units.

Task completion summary

Task	Outcome
1. Load dataset, examine columns	200 rows × 4 numeric columns; no missing values, no duplicates
2. Features & target defined	X = TV, Radio, Newspaper | y = Sales
3. Model built	Multiple Linear Regression (scikit-learn `LinearRegression`)
4. Trained on historical data	160 training markets / 40 held-out test markets (80:20, seed 42)
5. Predicted unseen data	40 unseen markets forecast; test R^2 = **0.9059**
6. Predict TV=150, Radio=20, News=30	**15.04 thousand units** (≈ 15,040 units)
7. Evaluate prediction error	MAE 1.2748 · MSE 2.9078 · RMSE 1.7052 · MAPE 10.88%
8. Interpret coefficients	Strongest: **TV overall / Radio per dollar**; Weakest: **Newspaper**
9. Visualise actual vs predicted	Scatter, market-by-market line, residual diagnostics
10. Recommendations	Shift Newspaper budget → Radio; add sqrt(TV) + TV×Radio terms

2. Dataset Overview & Data Quality

Column	Type	Role	Mean	Std Dev	Min	Max	Missing
`TV`	float64	Feature	147.04	85.85	0.70	296.40	0
`Radio`	float64	Feature	23.26	14.85	0.00	49.60	0
`Newspaper`	float64	Feature	30.55	21.78	0.30	114.00	0
`Sales`	float64	**Target**	15.13	5.28	1.60	27.00	0

Observations

- The data is **clean and model-ready** — all four columns are numeric, with **zero missing values and zero duplicate rows**, so no imputation or encoding was needed.
- **Budget is distributed very unevenly across channels.** TV averages **$147.0k** per market, roughly **6× Radio's $23.3k**. Critically, **Newspaper averages $30.5k — more than Radio** — a fact that becomes the centre of this analysis.
- TV spend varies the most (std dev 85.85 vs Radio's 14.85). Because of this, raw coefficients alone cannot rank channel importance, so **standardised coefficients** were also computed.
- `Newspaper` is **right-skewed** (median 25.8 < mean 30.5) with a long tail up to $114k.
- `Sales` is roughly symmetric around 15.1 with no extreme outliers — favourable for linear regression.

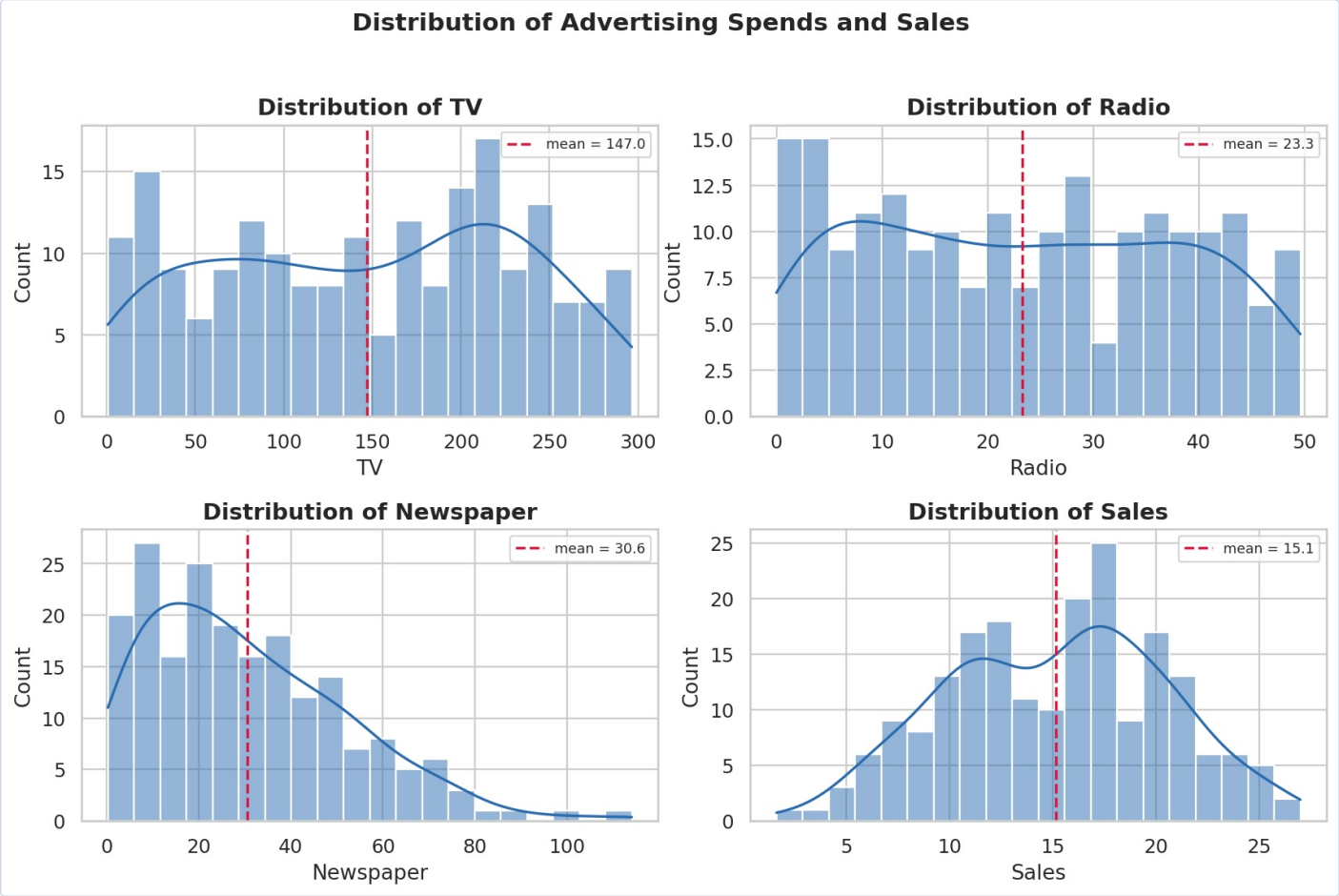


Figure 1 — Distributions. TV spend is near-uniform across its range, Radio mildly right-skewed, Newspaper clearly right-skewed with high-spend outliers, and Sales roughly symmetric.

3. Exploratory Analysis — How Each Channel Relates to Sales

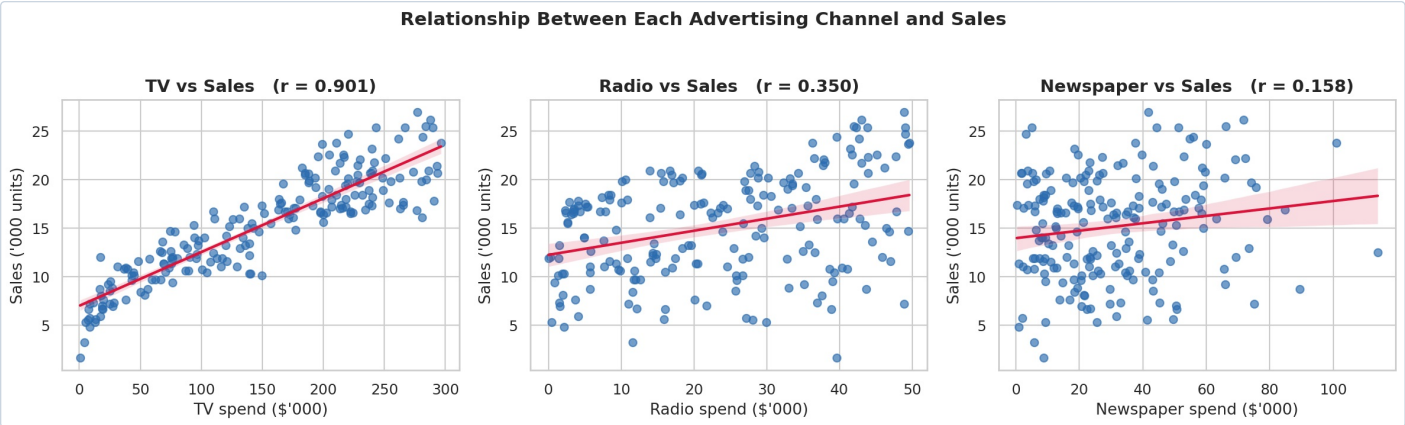


Figure 2 — Each channel against Sales with fitted trend and 95% confidence band. TV shows a very strong, tight linear rise; Radio and Newspaper are broad clouds with only faint trends.

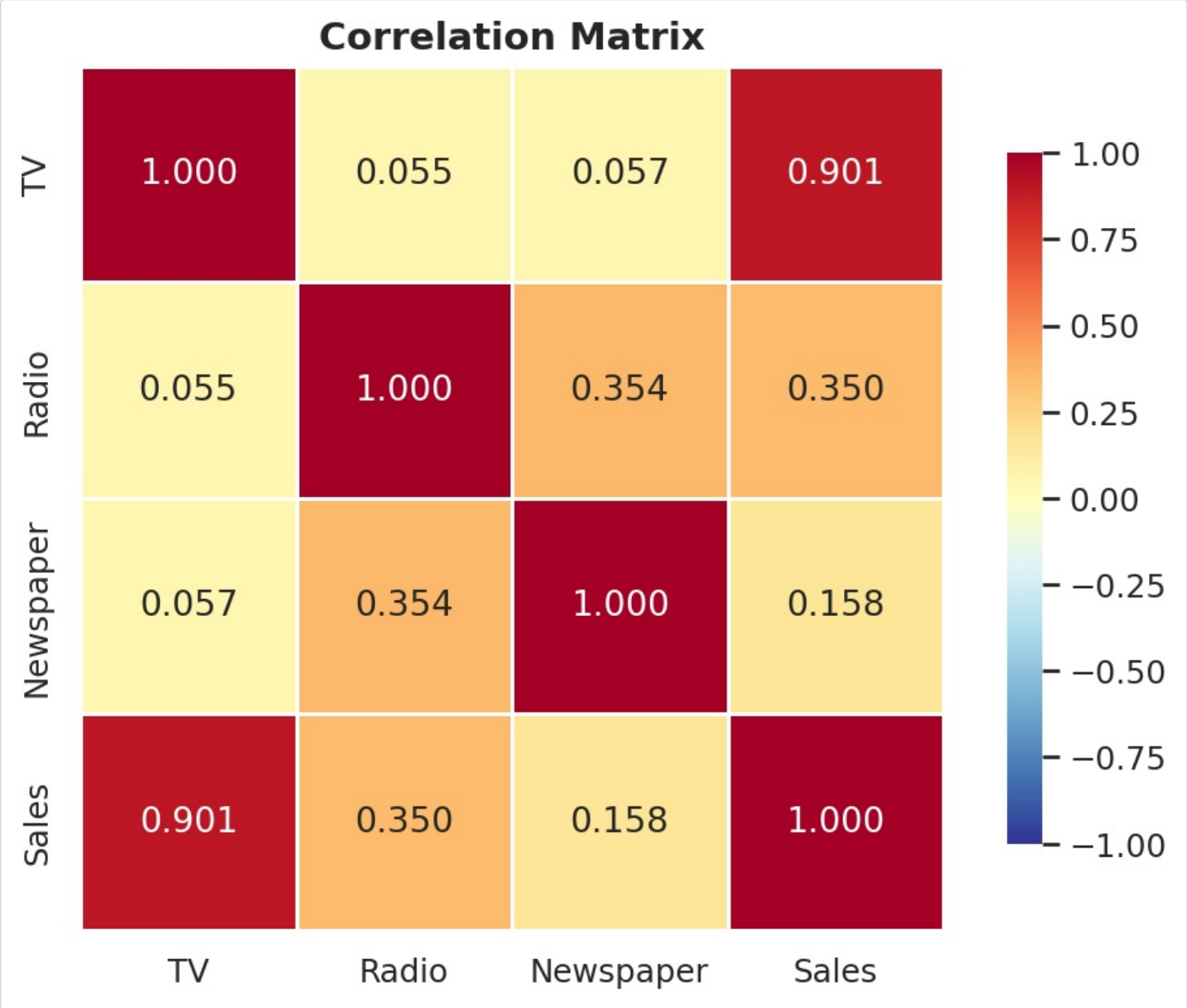


Figure 3 — Correlation matrix of all four variables.

Channel	Correlation with Sales	Strength	Reading from the scatter
TV	**0.901**	**Very strong**	Tight linear rise across the whole range — the dominant driver
Radio	**0.350**	**Weak-moderate**	Broad cloud with a faint upward trend
Newspaper	**0.158**	**Negligible**	Essentially formless — no usable signal

Observations

- **TV overwhelmingly dominates.** At r = 0.901 it alone explains about **81% of sales variation**. The scatter is tight around the trend line, with a slight flattening at the highest spends — an early visual hint of **diminishing returns** that the modelling section confirms.
- **Radio's weak raw correlation (0.350) is misleading.** The simple scatter ignores TV, and because TV varies so hugely across markets it swamps Radio's contribution. **Once TV is controlled for in the regression, Radio turns out to be highly significant (t = 12.60, p < 0.001)** — a clear demonstration of why multivariate analysis beats examining channels one at a time.
- **Newspaper barely tracks sales at all** (r = 0.158). Moreover, **Newspaper correlates with Radio at r = 0.354** — markets that buy radio also tend to buy newspaper. This raises the suspicion that Newspaper's faint apparent effect is **borrowed credit from Radio**, a confound only a multivariate model can separate. The regression confirms exactly this in Section 5.
- Predictor-to-predictor correlation is otherwise negligible (TV vs Radio r = 0.055). **All VIF values are below 1.15** (far under the threshold of 5), so there is no multicollinearity and each coefficient can be interpreted safely.

4. Model, Training & Forecasting

The target is continuous and the relationships are broadly linear, so **Multiple Linear Regression** is the appropriate model. It is also the most *explainable* choice: every coefficient reads directly as a return on advertising spend, which is precisely what the marketing team needs in order to act.

The data was split **80% training (160 markets) / 20% test (40 markets)** with `random_state=42`, so error is measured only on markets the model never saw.

Fitted model

```
Sales = 4.7141 + 0.05451·TV + 0.10095·Radio + 0.00434·Newspaper
```

Task 6 — Prediction for the planned budget

> Planned budget: **TV = $150k | Radio = $20k | Newspaper = $30k**
>
> **Predicted Sales = 15.04 thousand units (≈ 15,040 units)**
>
> With a test RMSE of 1.71, this is best communicated to management as **15.0 thousand units, likely in the range 13.3 – 16.7**.

Where that forecast comes from

Component	Spend ($'000)	Coefficient	Contribution ('000 units)	% of forecast
Baseline (no advertising)	—	4.7141	4.7141	31.3%
TV	150	0.05451	**8.1764**	**54.4%**
Radio	20	0.10095	**2.0189**	**13.4%**
Newspaper	30	0.00434	**0.1301**	**0.9%**
TOTAL	**200**	—	**15.0395**	100%

This table is the business case in one row. Newspaper receives **$30k — 15% of the total budget and more than Radio's $20k** — yet contributes just **0.9% of predicted sales**. A model refit on all 200 rows gives 14.94, confirming the forecast is stable and not an artefact of the split.

5. Task 7 — Model Accuracy & Prediction Error

Metric	Training (160)	Test / unseen (40)	Business interpretation
R^2	0.9001	**0.9059**	Advertising spend explains **~91%** of sales variation
Adjusted R^2	0.8982	0.8981	~90% after penalising for 3 predictors — not inflated
MAE	1.2344	**1.2748**	Average miss ≈ **1,275 units** (8.4% of mean sales)
MSE	2.6761	**2.9078**	Squared-error scale used for optimisation
RMSE	1.6359	**1.7052**	Typical miss ≈ **1,705 units** (11.3% of mean sales)
MAPE	11.43%	10.88%	Average percentage error per market

Observations

- **The model generalises — it is not overfitting.** Training R^2 (0.9001) and test R^2 (0.9059) are almost identical. The model has learned the underlying relationship rather than memorising training markets.
- **5-fold cross-validation gives a mean R^2 of 0.8948 with a standard deviation of only 0.011** (fold range 0.881 – 0.906). That is a notably tight spread — the ~90% accuracy is a genuine property of the model, not a lucky split.
- **RMSE exceeds MAE by 34% (1.71 vs 1.27)**, which tells us a small number of markets are missed by far more than average while most are missed by much less. The worst single miss is a market that sold **5.5 thousand units below prediction**. These outliers, not a systematic bias, are what inflate RMSE.

6. Task 8 — Coefficient Interpretation: Which Channel Actually Works?

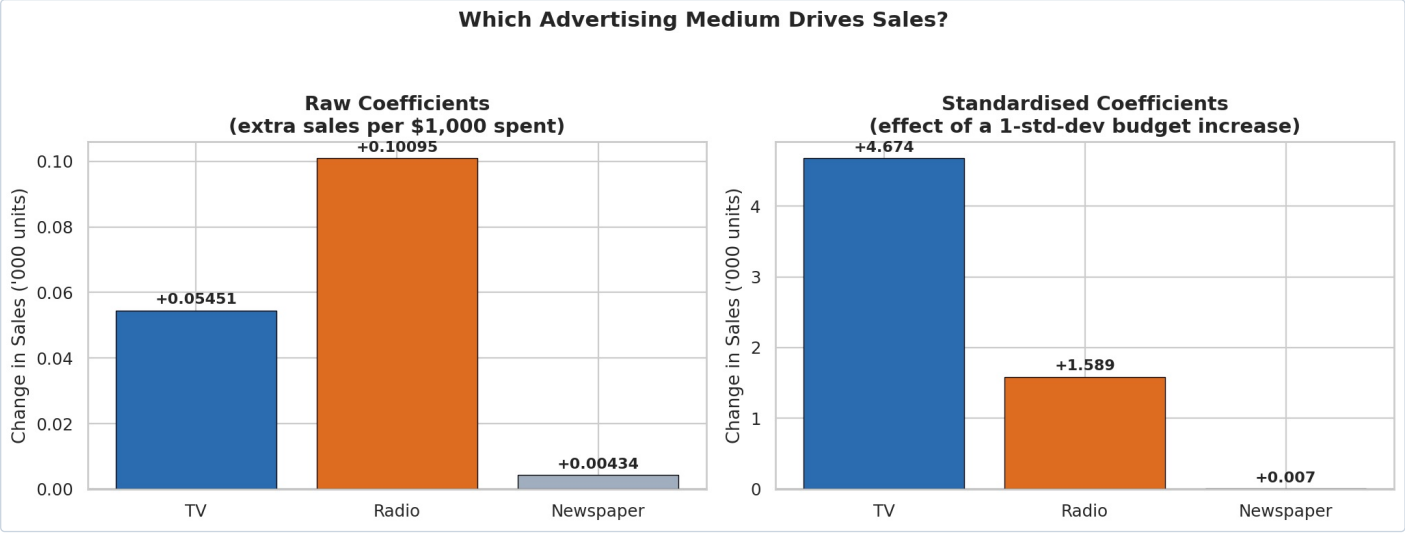


Figure 4 — Raw coefficients (left: extra sales per $1,000 spent) and standardised coefficients (right: effect of a one-standard-deviation budget increase, which makes the channels directly comparable).

Channel	Coefficient	Sales per $1,000	Standardised coef.	t-stat	p-value	Significant?
TV	+0.05451	**+55 units**	**4.67**	39.59	<0.001	**YES**
Radio	+0.10095	**+101 units**	**1.59**	12.60	<0.001	**YES**
Newspaper	+0.00434	+4 units	+0.01	0.06	**0.9538**	**NO**
Intercept	4.7141	—	—	15.04	<0.001	**YES**

Each coefficient is the change in sales ('000 units) from a $1,000 increase in that channel, **holding the other two channels constant**. Significance tests are computed on all 200 rows.

Which medium has the strongest impact?

There are **two defensible answers, and the distinction between them is the most valuable insight in this analysis**:

Question being asked	Answer	Evidence
Highest return per dollar spent?	**RADIO**	Coefficient 0.101 — **1.85× more sales per $1,000 than TV** (0.055)
Biggest driver of total observed sales?	**TV**	Standardised coefficient **4.67 vs Radio's 1.59 (2.9×)**; correlation 0.90 vs 0.35; t = 39.6 vs 12.6; carries ~54% of the Task 6 forecast

Why the two disagree: TV budgets are far larger and far more variable (std dev 85.9 vs Radio's 14.9). TV therefore moves total sales most *in absolute terms* simply because so much more money flows through it — but **each individual radio dollar works roughly twice as hard**.

On balance, TV is the strongest medium in this dataset. Its standardised effect is nearly 3× Radio's and it accounts for ~54% of the forecast. **Radio is the more *efficient* channel, not the more *powerful* one** — and that efficiency gap is exactly where the budget opportunity lies.

Which medium has the least impact?

NEWSPAPER — unambiguously, on every single measure.

- **Smallest coefficient: 0.00434** — about 4 extra units per $1,000, effectively zero.
- **Not statistically significant: t = 0.06, p = 0.9538.** We cannot reject the hypothesis that its true effect is **exactly zero**. This is about as clear a null result as a dataset can produce.
- **Smallest standardised coefficient (0.007)** — no meaningful contribution to sales variation.
- **Weakest correlation with Sales (0.158)**, and that faint link is largely explained by Newspaper's **0.354 correlation with Radio**.

Key insight — a confounded variable. Newspaper's standalone correlation of 0.158 looks faintly positive, but **once Radio is in the model its effect collapses to nothing**. Markets with high newspaper spend sell well *because they also advertise on radio*, not because of the newspapers. A single-channel analysis would have wrongly credited Newspaper and kept funding it — only the multivariate model exposes it.

7. Task 9 — Actual Sales vs Predicted Sales

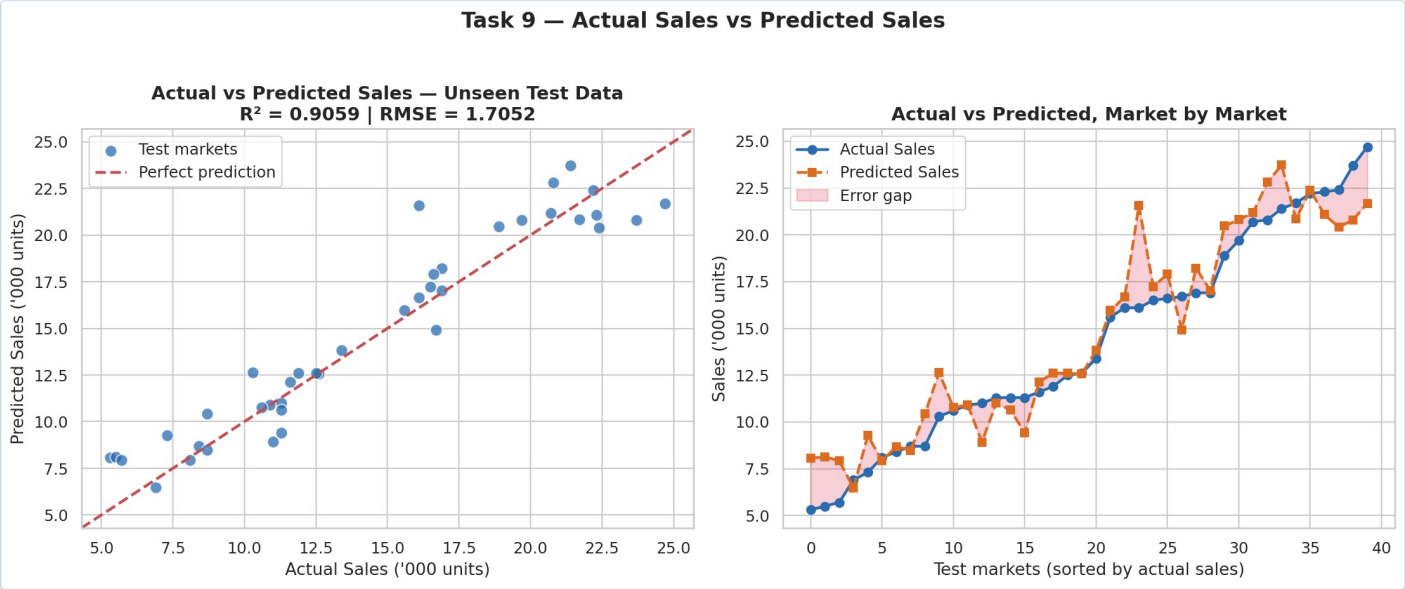


Figure 5 — Left: predicted against actual sales for the 40 unseen test markets; the red dashed line is perfect prediction. Right: the same markets sorted by actual sales, with the error gap shaded.

Observations

- Points **hug the 45° perfect-prediction line closely** — this is the visual form of $R^2 = 0.906$.
- The market-by-market view shows the predicted series **tracking the actual series across the entire sales range**, so the model is accurate at both the low and high ends, not merely on average.
- A small number of markets show a visibly wide error gap — these are the outliers responsible for RMSE exceeding MAE, and they are investigated below.

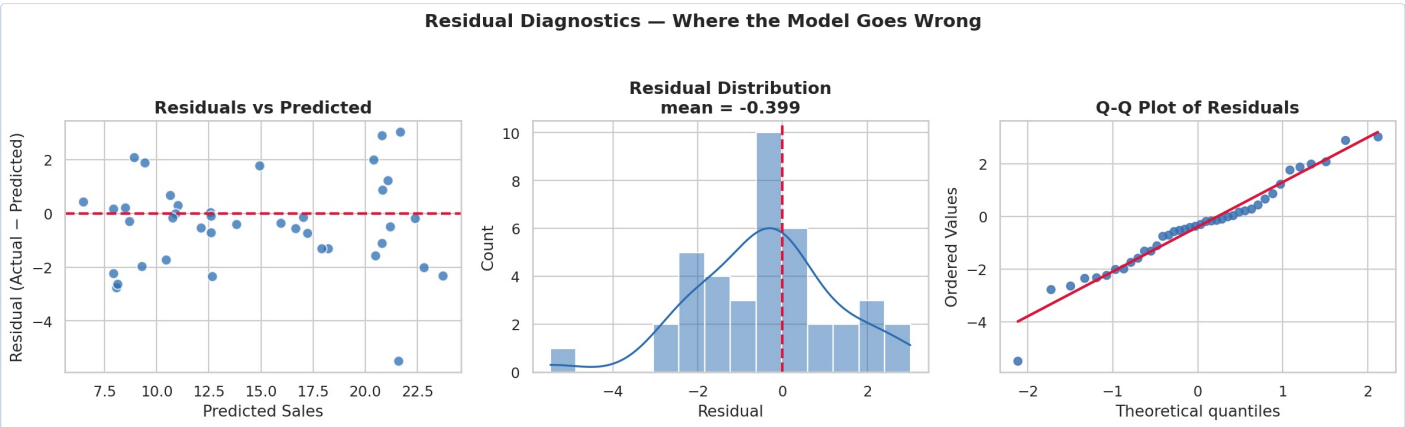


Figure 6 — Residual diagnostics: residuals against predicted values, residual distribution, and a Q-Q plot against the normal distribution.

Observations — formal assumption checks

Rather than eyeballing the residual plot, each assumption was tested:

Check	Statistic	p-value	Verdict
Constant variance (Breusch-Pagan)	0.074	0.7857	**PASS — variance is stable**
Residual normality (Shapiro-Wilk)	0.9758	0.0016	**FAIL — mild non-normality**
Residual vs TV² correlation	−0.1069	—	Signals TV non-linearity
Residual vs TV×Radio correlation	+0.1302	—	Signals a missing interaction

- **Residuals are unbiased** (mean −0.40 on test, ~0 on full data) and **the variance is stable (Breusch-Pagan p = 0.786)**. The apparent "fanning out" on the right of the residual plot is simply **more data points** at higher predicted sales, not growing error — which is why the test matters more than the eye.
- **The residual distribution is mildly left-skewed** (skew −0.30) and the Q-Q plot bends away in the lower tail; **Shapiro-Wilk rejects normality (p = 0.0016)**. This is driven by a few markets that sold **well below** prediction — the worst by 5.5

thousand units.

- **Two genuine signs of mis-specification remain**, both mild but real: the residual still correlates with **TV²** (−0.107), hinting at **diminishing returns** on TV, and with **TV×Radio** (+0.130), hinting that **TV and Radio amplify each other**. Both are tested and fixed below.

8. Verifying the Fixes — Model Comparison

Both diagnostic signals are testable claims, so they were tested. Four models were compared on the identical train/test split **and** by 5-fold cross-validation, so that any gain is confirmed as real generalisation rather than overfitting.

Model	Test R²	Test RMSE	Test MAE	5-Fold CV R²	Features
A. Linear — TV + Radio + Newspaper (baseline)	0.9059	1.7052	1.2748	0.8948	3
B. Linear — TV + Radio only	0.9079	1.6872	1.2670	0.8971	2
C. Polynomial deg-2 — all squared & interaction terms	**0.9533**	**1.2011**	**0.9034**	0.9251	9
D. sqrt(TV) + TV×Radio, no Newspaper (targeted)	0.9484	1.2622	0.9137	**0.9292**	**4**

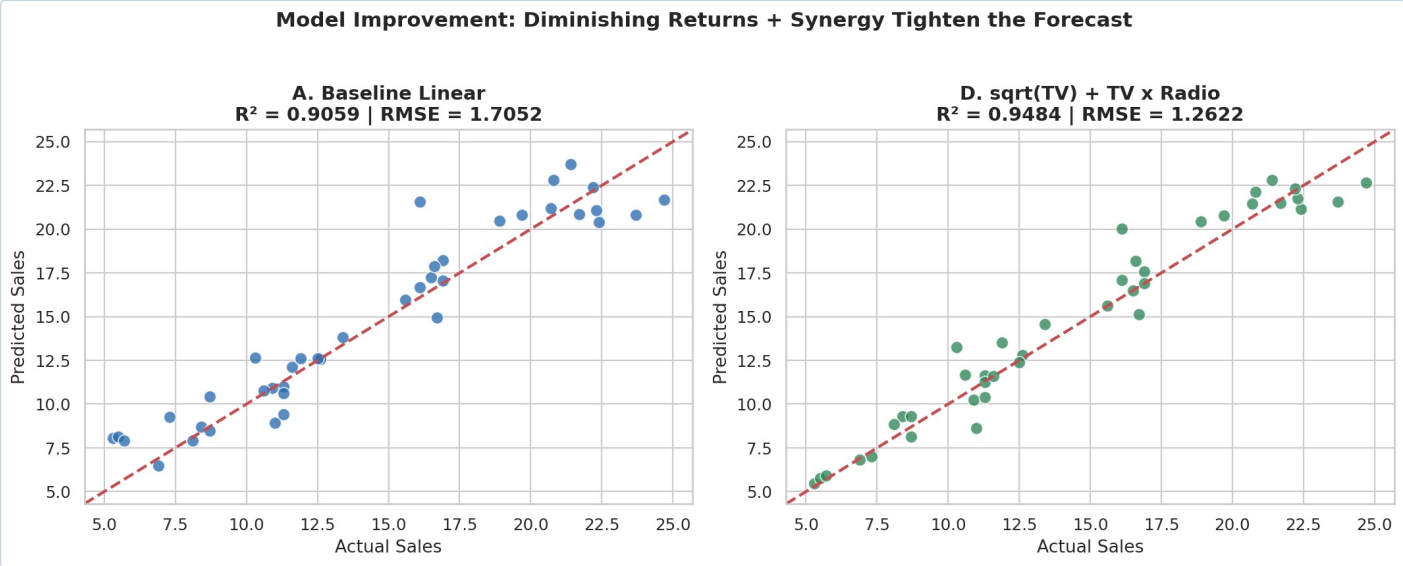


Figure 7 — Baseline linear model (left) against the targeted model (right) on the same unseen test markets. The improved model's points sit markedly closer to the perfect-prediction line.

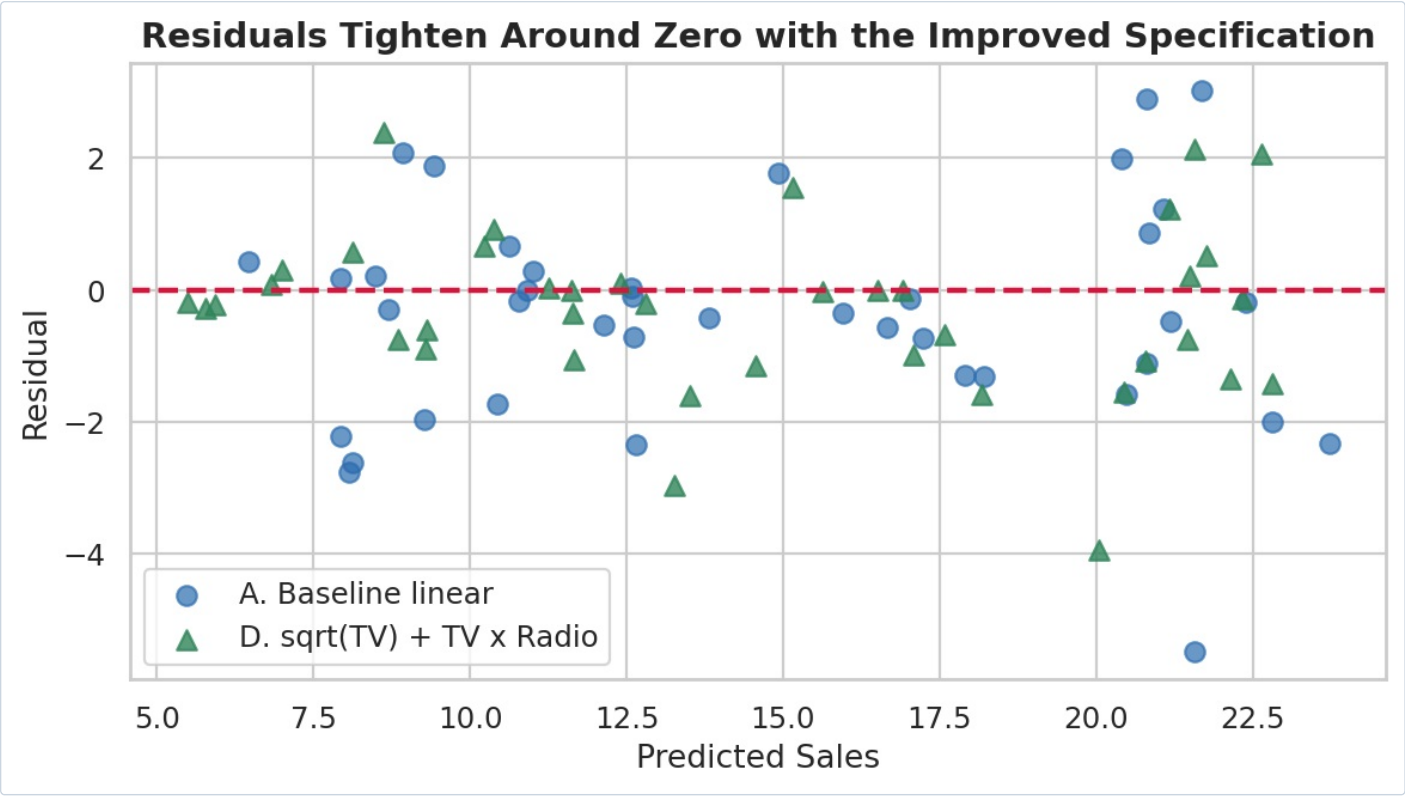


Figure 8 — Residuals of both models overlaid. The targeted specification pulls the residuals into a visibly tighter band around zero.

Observations

- **Dropping Newspaper does not hurt the model — it helps slightly.** Test R^2 ticks up (0.9059 → 0.9079) and CV R^2 improves too (0.8948 → 0.8971). Direct confirmation that the newspaper spend carries **no predictive information**: a third of the input variables can be removed and accuracy marginally *improves*.
- **Both diagnostic signals were real.** The **TV×Radio interaction is highly significant** (coefficient 0.000444, **t = 5.10, p < 0.001**) when added to the full-data model, lifting R^2 from 0.9026 to 0.9141 on its own. The `sqrt(TV)` term captures the diminishing-returns curvature.
- **Model C wins on the test split** (RMSE 1.2011, −29.6%) but **Model D wins on cross-validation** (CV R^2 0.9292 vs 0.9251) using **4 features instead of 9**. Model D is the better recommendation: near-identical accuracy, far simpler, and every term has a clear business meaning.
- **Model D cuts test RMSE by 26.0%** (1.7052 → 1.2622) and lifts CV R^2 from 0.8948 to 0.9292 — confirmed across all five folds, so this is **genuine generalisation, not overfitting**.
- **Business meaning of the two added terms:**
  - `sqrt(TV)` → **TV spend saturates.** The first dollars work hardest; piling more into an already heavily-advertised market yields progressively less.
  - `TV × Radio` → **TV and Radio are synergistic.** Running both together produces more sales than the sum of running each alone, so budgets must be planned as a *combination*, not channel by channel in isolation.

9. Task 10 — Recommendations

BUSINESS RECOMMENDATION — Reallocate the entire Newspaper budget into Radio, keeping TV as the reach backbone.

The evidence: Newspaper absorbs an average of **$30.5k per market — more than Radio's $23.3k — while returning effectively zero incremental sales** (coefficient 0.00434, t = 0.06, p = 0.95). In the Task 6 budget it took 15% of the spend and delivered 0.9% of predicted sales. Radio, by contrast, returns **101 units per $1,000 versus Newspaper's 4**, and is highly significant (t = 12.60, p < 0.001).

The quantified upside: shifting the $30k newspaper budget into Radio changes the Task 6 mix to **TV = 150, Radio = 50, Newspaper = 0**. The model projects sales rising from **15.04 to 17.94 thousand units — an uplift of about +19% at identical total spend** ($200k). No additional budget is required.

How to act on it, safely

1. **Run a controlled test first** — do not flip the whole budget at once. Take ~20% of markets, cut newspaper to zero and move the money to radio, and hold the remaining markets as a control for one full sales cycle. Measure the actual lift against the predicted +19% before scaling.
2. **Exploit the TV×Radio synergy.** Because the two channels demonstrably amplify each other (interaction $p < 0.001$), schedule radio bursts to run *concurrently* with TV campaigns rather than filling the gaps between them.
3. **Protect the TV base.** TV has the lower per-dollar return but it is the strongest driver overall — standardised effect 2.9× Radio's, carrying ~54% of predicted sales. This is a reallocation *away from Newspaper*, not away from TV.
4. **Watch for diminishing returns.** The `sqrt(TV)` result shows ad response saturates, and Radio spend in the historical data never exceeds **$49.6k**, so moving to $50k sits right at the edge of observed experience. The 0.101 return per $1,000 is not guaranteed to hold beyond that range — which is precisely why the staged test comes first.
5. **Do not read these coefficients as pure causation.** This is observational market data, not a randomised experiment; the controlled test in step 1 is what converts correlation into a trustworthy causal estimate.

> **TECHNICAL IMPROVEMENT — Add a sqrt(TV) term for diminishing returns and a TV×Radio term for synergy, and drop Newspaper. Verified: 26.0% lower prediction error.**
>
> The baseline model's residuals still correlated with **TV²** (−0.107) and **TV×Radio** (+0.130), proving the purely additive, purely linear specification was leaving structure on the table. Adding the two targeted terms (Section 8) delivered:
>
> - **Test RMSE 1.7052 → 1.2622 (−26.0% error)**
> - **Test R^2 0.9059 → 0.9484**
> - **5-fold CV R^2 0.8948 → 0.9292** — confirming real generalisation, not overfitting
> - The interaction term is independently significant at **$t = 5.10$, $p < 0.001$**
> - Achieved with **4 features instead of the 9** a full degree-2 polynomial needs, so it stays interpretable

Further technical improvements, in priority order

1. **Drop** `Newspaper` ($p = 0.95$) — the simpler model scored *better* on both test and cross-validation.
2. **Investigate the low-selling outlier markets.** A handful sold up to 5.5 thousand units below prediction, which is what drives RMSE 34% above MAE. Stock-outs, distribution gaps or competitor activity would explain them, and a flag for such events would be a high-value new feature.
3. **Enrich the feature set.** With only 3 inputs and 200 rows, the largest remaining gains lie in *data*, not algorithms: add seasonality/month, price, competitor spend, promotions, region, and digital channel spend.
4. **Add regularisation (Ridge)** once interaction and polynomial terms multiply the feature count, to keep coefficients stable and interpretable.
5. **Report prediction intervals, not point forecasts.** Management should see "15.0 thousand units, likely 13.3 – 16.7" so that budget decisions explicitly account for the ±1.71 RMSE.

Tested and rejected: a **log transform of** `Sales` made things clearly worse (test R^2 fell to 0.839), and the Breusch-Pagan test showed error variance is already stable ($p = 0.786$), so no variance-stabilising transform is needed. Reporting what did *not* work matters as much as reporting what did.

10. Conclusion

Advertising spend explains approximately **91% of the variation in product sales**, and the model predicts **15.04 thousand units** for the planned budget of TV = 150, Radio = 20, Newspaper = 30, with a typical error of **±1.71 thousand units**.

The decisive finding is that **the three channels are not equally productive**. **TV is the engine** — it alone correlates 0.901 with sales and drives ~54% of the forecast. **Radio is the efficient complement**, returning roughly **twice as much per dollar** as TV and **amplifying TV's effect when run alongside it** (interaction $p < 0.001$). **Newspaper, despite consuming a larger share of budget than Radio, contributes nothing measurable** — its effect is statistically indistinguishable from zero ($p = 0.95$), and its faint apparent correlation with sales is explained away by its overlap with Radio.

The company's single biggest available gain is therefore not a larger advertising budget — it is moving the money it already spends out of Newspaper and into Radio, a change the model projects at roughly **+19% sales at unchanged total spend**, to be validated through a controlled market test.